

Requirements Traceability Matrix

Module: pqc_crypto_module v0.1.0 **Date:** 2026-04-28

Disclaimer: This package does not imply FIPS 140-3 validation or CMVP certification.

1. Purpose

This matrix maps each security requirement to its implementation file, test file, documentation file, and current status. It provides traceability from requirement through code to evidence.

2. Traceability Table

#	Requirement	Implementation File(s)	Test File(s)	Documentation
R-01	Approved mode only uses FIPS-approved algorithms	crates/ pqc_crypto_module/ src/lib.rs, src/ state.rs, approved_mode.rs	crates/ pqc_crypto_module/src/ self_test.rs, fips_readiness.rs, unit tests in state.rs	FIPS_140_3_#3, #4
R-02	Self-tests must pass before any crypto operation	crates/ pqc_crypto_module/ src/self_tests.rs, src/approved_mode.rs	self_tests.rs (KAT tests), fips_readiness.rs, tools/ acvp_dry_run/ (15 ACVP- inspired vectors)	MOCK_AUDIT_#7 (CLOSED), FIPS_140_3_#7
R-03	Error state is fail-closed	crates/ pqc_crypto_module/ src/state.rs, src/ errors.rs, approved_mode.rs	state.rs (error state tests), fips_readiness.rs (error_state_rejects_all)	MOCK_AUDIT_#7
R-04	ML-DSA-65 sign and verify	crates/ pqc_crypto_module/ src/mldsa.rs	self_test.rs (sign/verify/ corrupt KAT), no_fallback.rs	ACVP_DRY_RUN_#7 Section 3
R-05				

#	Requirement	Implementation File(s)	Test File(s)	Documentation
R-06	ML-KEM-768 encapsulate and decapsulate	crates/ pqc_crypto_module/ src/mlkem.rs	mlkem.rs (8 unit tests), self_tests.rs (KAT with roundtrip + invalid ct), api_boundary.rs, tools/ acvp_dry_run/ (5 encapsThenDecaps vectors)	MOCK_AUDIT_ (CLOSED)
	SHA3-256 hashing	crates/ pqc_crypto_module/ src/hashing.rs	self_test.rs (SHA3 KAT), no_fallback.rs	ACVP_DRY_RUN Section 5
	Zeroization of private keys and shared secrets	crates/ pqc_crypto_module/ src/types.rs (key types with ZeroizeOnDrop)	key_zeroization.rs	MOCK_AUDIT_ F-005, FIPS_140_3_ #10
	Legacy algorithms blocked in approved mode	crates/ pqc_crypto_module/ src/legacy.rs (ensure_not_approved() guard), src/identity/	approved_vs_legacy.rs, crypto_boundary.rs	MOCK_AUDIT_ FIPS_140_3_ #4
	No raw crypto operations outside module boundary	All 189 source files outside crates/ pqc_crypto_module/ src/	tests/crypto_boundary.rs (189/189 clean, 5 tests)	MOCK_AUDIT_ FIPS_140_3_ #2
	Reproducible build	scripts/ clean_room_build.sh, Cargo.lock (pinned)	CI pre-lab-audit.yml, manual execution (build comparison)	CLEAN_ROOM
	RNG failure results in operation failure, no fallback	crates/ pqc_crypto_module/ src/rng.rs	self_test.rs (continuous RNG test), rng.rs unit tests	ENTROPY_RNG Section 4
R-12	Cryptographic boundary enforced (11 source files)	crates/ pqc_crypto_module/ src/ (11 files), crate isolation	tests/crypto_boundary.rs (5 tests)	MOCK_AUDIT_ FIPS_140_3_ #2

3. Status Summary

Status	Count	Requirements
PASS	8	R-02, R-03, R-04, R-05, R-06, R-09, R-11, R-12
PARTIAL	4	R-01, R-07, R-08, R-10
FAIL	0	–

4. Blocking Items for Lab Intake

Requirement	Finding	Blocking?
R-05 (ML-KEM)	F-001	CLOSED – pqcrypto-mlkem implementation with roundtrip verification
R-02 (Self-tests / ACVP)	F-002	CLOSED / PARTIAL-OFFICIAL – ACVP dry-run harness functional
R-10 (Reproducible build)	F-011	Yes – lifecycle assurance requires verified hash comparison
R-01 (Approved mode)	F-006	Medium – FSM enforcement gap at some call sites
R-08 (Legacy blocked)	F-006	Medium – same root cause as R-01

5. Cross-References

- Finding IDs (F-xxx) reference `MOCK_AUDIT_REPORT.md`
- IG checklist items reference `FIPS_140_3_IG_CHECKLIST.md`
- All findings aggregated in `FINDINGS_REGISTER.md`

End of traceability matrix.